

7. Output Devices (Output Yantra)

Output devices woh hardware hain jo processed data (jo information ban chuka hai) ko hamein dikhate, sunate ya print karte hain.

7.1 Monitors (Screen / Pardaa)

Monitor sabse common output device hai. Iska quality "pixels" se decide hota hai. Pixel ek chhoti si dot hoti hai, laakhon pixels milkar image banate hain.

Resolution matlab screen par kitne pixels hain (e.g., 1920x1080 matlab 1920 columns, 1080 rows).

Type	Full Form	Kaam Kaise Karta Hai	Features
LCD	Liquid Crystal Display	Liquid crystals ek backlight ke saamne shutter ki tarah open/close hote hain.	Purani technology, sasti, thick, average picture quality.
LED	Light Emitting Diode	Actually yah bhi LCD hi hai, bas backlight CCFL lamp ki jagah LED bulbs hoti hai.	LCD se patli, zyada efficient, better colors. (Aaj kal sab yahi use hai).
OLED	Organic Light Emitting Diode	Isme har pixel apni khud ki light paida karta hai. Backlight ki zaroorat hi nahi.	Sabse patli, True Black (pixel off=andhera), amazing contrast, expensive. High-end phones/TV.

Common Ports for Monitor: VGA (purana, analog), HDMI (modern, audio+video), DisplayPort (high refresh rate gaming).

7.2 Printers (Mudrak Yantra)

Digital file se physical paper copy banata hai.

Type	Kaam Kaise	Quality	Cost	Best For
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Type	Karta Hai	Quality	Cost	Best For
Inkjet	Paper par chhoti-chhoti ink ki boondein spray karta hai.	Bahut acchi (Photos)	Printer sasta, ink mehngi	Ghar/Office ke color documents.
Laser	Static electricity aur heat se fine powder (toner) ko paper par chipkata hai. Jaise photocopy machine.	Text ke liye sharp	Printer mehnga, per page cost sasti	Office ke black & white documents (High Volume).
Plotter	Pen ya marker use karke bade paper par draw karta hai.	Very high, large scale	Bohot mehnga	Engineers, architects ke maps, blueprints.

7.3 Other Output Devices

- **Speaker:** Digital signal ko sound waves mein badalta hai.
- **Projector:** Ek screen ki jagah, image ko bade screen/diwar par project karta hai. Classroom, presentations.

 Topic 7: Summary

Device	Output Type	Best Example Use
OLED Screen	Visual	iPhone, Premium TV (Best contrast).
Laser Printer	Printed Text	Office ke 1000-pages ke documents.
Inkjet Printer	Printed Photo	Ghar pe photo print karna.
Plotter	Large Print	Building ka map/blueprint.

8. Operating System (OS) — Computer Ki Aatma

Definition: Operating System ek **system software** hai jo computer hardware

Definition: Operating system ek **system software** hai jo computer hardware aur user ke beech **interface** (madhyam) ka kaam karta hai. Bina OS ke computer ek dabba hai.

Real-Life Analogy: Ek **office manager**. Office ke resources (Desk=RAM, Almirah=Storage, Employees=CPU) ko manage karta hai aur visitor (User) aur staff (Hardware) ke beech mein baat karwata hai.

8.1 Core Functions of OS

1. **Process Management:** Ek saath MS Word, Chrome, aur gaane chal rahe hain — OS decide karta hai CPU kisko kitna time milega.
2. **Memory Management:** Kaunse program ko RAM mein kitni jagah milegi, ye OS karta hai.
3. **File Management:** Aapki files folder mein kaise organize hongy, ye OS ka file system karta hai (e.g., NTFS, FAT32).
4. **Device Management:** Printer, mouse, keyboard se baat karna.
5. **Security:** Password, user accounts, unauthorized access se bachana.

8.2 Important OS Concepts

- **Kernel:** Operating system ka **heart/dil**. Ye hardware (CPU, RAM) se directly baat karta hai. Sabse powerful core component.
- **Shell:** OS ka **outer layer**. User commands leta hai (mouse click, keyboard) aur kernel tak pahunchata hai. **GUI (Graphical User Interface)** aur **CLI (Command Line Interface)** dono shell ke types hain.
- **Process vs Thread:**
 - **Process:** Ek executable program. Jaise MS Word ek process hai.
 - **Thread:** Process ke andar chhota worker. MS Word mein ek thread typing kar raha hai, doosra spell-check kar raha hai.
- **Multitasking:** Ek saath kayi apps chalana. (CPU tezi se ek se doosre app mein switch karta hai).
- **Virtual Memory:** Jab RAM full ho jaye, to OS hard disk ke ek part ko "nakli RAM" ki tarah use karta hai. Isse apps chalte rahte hain, par slow ho jaata hai.
- **Paging:** Virtual memory mein data chhote-chhote blocks (pages) mein divide hota hai.

8.3 Types of Operating Systems

OS	Developer	Key Feature	Use Case
		Most popular version	

Windows	Microsoft	Most popular, user-friendly, gaming ke liye best.	Personal computers, offices.
macOS	Apple	Smooth, secure, great for creativity (video/photo editing).	Apple MacBooks, iMacs.
Linux	Open Source	Free, highly customizable, bahut secure.	Servers (90% cloud), programmers, hackers. (e.g., Ubuntu, Fedora)
Android	Google	Linux-based, mobile ke liye modified.	Smartphones (Samsung, OnePlus).
iOS	Apple	Apple mobile devices ke liye. Highly secure, closed ecosystem.	iPhones, iPads.

✓ Topic 8: Summary

Key Term	Simple Meaning
OS	Hardware-User interface manager.
Kernel	Heart of OS, talks directly to hardware.
Shell	Outer layer, takes user commands.
Process	An application in execution.
Thread	A small worker inside a process.
Virtual Memory	RAM full hone par HDD/SSD ko RAM jaisa use karna.

9. Networking (Computeron Ka Jaal)

Jab do ya zyada computers aapas mein data share karne ke liye judte hain, use networking kehte hain.

9.1 Types of Networks (by Size)

- **LAN (Local Area Network):** Chhota network. Ek building ya campus. Ghar ka Wi-Fi, school lab. Fast, cheap.
- **MAN (Metropolitan Area Network):** Ek city ke andar network. Cable TV network. LAN se bada, WAN se chhota.
- **WAN (Wide Area Network):** Duniya ka sabse bada network — **Internet**. Desh, mahadesh sabko jodta hai.

9.2 Networking Devices

Device	Kaam (Function)	OSI Layer	Analogy
Hub	Data aaya, sabhi devices ko bhej diya (stupid device).	1 (Physical)	Ek aadmi jo chilla kar sabko sunata hai.
Switch	Data sirf usi device ko bhejta hai jiske liye hai (smart device). MAC address se pehchan karta hai.	2 (Data Link)	Postman jo ghar ka address dekh kar letter daalta hai.
Router	Do alag-alag networks ko jodta hai. Aapke ghar ke network ko Internet se. IP address use karta hai.	3 (Network)	Aapke society ka main gate, jahan se baahar ki duniya mein jaate hain.
Modem	Telephone line ke Analog signal ko computer ke Digital signal mein badalta hai.	-	Translator. Jo ghar ka broadband deta hai woh usually Modem+Router combo hota hai.
Gateway	Do completely different protocol ke networks ko jodta hai.	-	Protocol translator.
Firewall	Network ka security guard . Incoming/outgoing		

Firewall	traffic check karta hai, rules ke hisaab se allow ya block.	-	Security guard.
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9.3 Important Protocols (Niyam)

Protocol matlab rules ka set jisse devices aapas mein baat kar sakein.

Protocol	Full Form	Port	Kaam	Analogy
TCP	Transmission Control Protocol	(varies)	Bharosemand (Reliable). Data bhejne se pehle connection banata hai, guarantee karta hai data deliver hoga. WhatsApp message.	Registered Post. Receipt aati hai.
UDP	User Datagram Protocol	(varies)	Tez (Fast), par bharosemand nahi. Data bhej diya, pahuncha ya nahi, check nahi karta. Live video streaming, game.	Saada Daak. Bhej diya, kho gaya to gum.
HTTP	HyperText Transfer Protocol	80	Website ka data (unsecured) transfer karta hai.	Khula patra.
HTTPS	HTTP Secure	443	HTTP + SSL Encryption. Website ka data secure transfer karta hai. Banking, shopping.	Band lifafa, muhar lagaya hua.
FTD	File Transfer	21	Server se files download/upload	Courier

Protocol	Port	Download/Upload	Service
SMTP	Simple Mail Transfer Protocol 25	Email bhejne ke liye.	Post office ka "outbox".
POP3 / IMAP	Post Office Protocol 3 / Internet Message Access Protocol 110 / 143	Email receive karne ke liye. POP3 download karta hai, IMAP server par hi rakh kar phone/laptop se sync karta hai.	Post office ka "inbox".
DNS	Domain Name System 53	Domain name (like <code>www.google.com</code>) ko IP address (like <code>142.250.1.1</code>) mein badalta hai. Internet ki phonebook.	Phonebook Truecaller.
DHCP	Dynamic Host Configuration Protocol 67/68	Aapke device ko automatically ek IP address assign karta hai jab aap Wi-Fi se judte hain.	Receptionist jo aate hi room ki chabhi deta hai.
SSL/TLS	Secure Sockets Layer/Transport Layer Security -	Data ko encrypt karta hai client-server ke beech. HTTPS ka 'S' inhi se aata hai.	Muhar band lifafe ki suraksha.

9.4 Addressing

- **IP Address (Internet Protocol):** Aapke device ka online pata. Jaise `192.168.1.5` (IPv4) ya naya `2001:db8::1` (IPv6).
- **MAC Address (Media Access Control):** Aapke device ka physical,

permanent address. Factory set hota hai. Jaise `00:1A:2B:3C:4D:5E` .

- **Difference:** MAC address aapka ghar ka address hai (nahi badalta), IP address hotel ka room number hai (roz badal sakta hai).

9.5 Modern Protocols (QUIC, HTTP/2, HTTP/3)

- **HTTP/2:** Google ke SPDY par based. Ek hi connection mein multiple requests/responses, multiplexing, faster page loading.
- **HTTP/3:** Ye ab TCP nahi, **QUIC** protocol use karta hai jo UDP par based hai. Connection time zero. Jis jagah network switch karte hain (Wi-Fi to mobile data) connection toot'ta nahi. YouTube, Facebook isi use karte hain.

✅ Topic 9: Summary

Concept	Key Detail
LAN/MAN/WAN	Size-based network types.
Hub/Switch/Router	Dumb sharer / Smart MAC sender / Inter-network IP router.
TCP vs UDP	Reliable (Registered Post) vs Fast (Normal Post).
HTTP vs HTTPS	Unsecured vs Secured (SSL/TLS).
DNS	Converts domain name to IP address (Phonebook).
QUIC/HTTP3	New gen, faster, built on UDP.

10. Internet (Antarjaal)

Internet **network of networks** hai. Pure vishwa ke computers ka ek jaal.

10.1 Key Terms

Term	Meaning	Example
Website	Web pages ka collection ek domain ke under.	<code>wikipedia.org</code>
Domain Name	Website ka naam, padhne mein aasan.	<code>amazon.in</code>

Hosting	Website ki files ko 24x7 internet se jude server par rakhna.	Ghar nahi, online almirah.
Browser	Website dekhne ka software.	Chrome, Firefox, Safari.
Search Engine	Internet par website dhundhne ka tool.	Google, Bing.
URL	Uniform Resource Locator. Poora web address.	<code>https://www.google.com/search?q=computer</code>
Cookies	Website aapke browser mein chhoti file save karti hai (login info, preferences).	Server ki yaaddaasht.
Cache	Website ka kuch data (images) browser save kar leta hai, agle baar jaldi load hota hai.	RAM ki tarah temporary memory.
Session	Jab aap kisi website par logged in rehte ho, woh period.	Login se logout tak ka samay.

Topic 10: Summary

Term	Analogy
Domain	Ghar ka naamplate.
Hosting	Ghar ki zameen/plot.
Browser	Ghar ki khidki jisse aap baahar dekh sakte hain.
Cookies	Dukaan daara aapki pasand ka record rakhna.

Cache	Bar-bar kaam aane wali cheez haath ke paas rakhna.
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11. Software (Computer Ke Program)

Hardware ko chalanewala jadoo.

11.1 Types of Software

1. **System Software:** Background mein chalta hai, hardware control karta hai.
 - **Operating System:** (Windows, Linux)
 - **Device Drivers:** Hardware aur OS ke beech translator. Jaise printer ka driver.
 - **Utility Software:** Maintenance tools. (Antivirus, Disk Defragmenter)
2. **Application Software:** Jo aap directly use karte hain.
 - General Purpose (MS Word, Excel, Photoshop)
 - Specific Purpose (Hotel booking, Hospital management software)

11.2 Language Translators (Code ko Machine Bhasha Mein Badalna)

Tool	Kaam	Analogy
Compiler	Poora code ek saath scan karta hai, error check karta hai, fir pure program ki file banata hai (.exe). Ek baar bana, fir baar-baar chalao.	Poori kitaab ka ek saath anuvaad.
Interpreter	Code line-by-line padhta aur turant execute karta hai. Error mila to wahin ruk gaya. Hamesha interpreting ki zaroorat.	Ek saath ek line ka anuvaad, bina likhe.
Assembler	Assembly language (low-level) code ko machine code mein badalta hai.	-

12. Programming Basics (Programming Ki Neev)

Kisi problem ko computer se solve karwane ki step-by-step recipe.

12.1 Key Concepts

Term	Meaning	Analogy
Algorithm	Kisi kaam ko karne ka step-by-step tareeka.	Chai banane ki recipe.
Flowchart	Algorithm ko diagram ke roop mein dikhana. Shapes se process batana.	Recipe ki photo guide.
Variable	Ek dabba jisme aap value store kar sakte hain (badal sakti hai).	Score = 0. (Game khelte waqt badalega).
Constant	Ek dabba jiski value fix hai.	Pi = 3.14159.
Data Types	Variable mein kis type ka data hai (Number, Text, True/False).	Integer (5), String ("Shivani"), Boolean (Yes/No).
Loop	Ek kaam ko baar-baar karna jab tak condition true hai.	Jab tak plate mein roti hai, khao.
Function	Code ka ek chhota reusable block.	Dadhi banane ka function. Jaroorat padi, bula liya.
Array	Ek list, ek hi type ke kai variables ek saath.	Class ke bachchon ke roll numbers ki list.
Pointer	Variable ka direct memory address.	Ghar ka pura physical address.
OOP	Object Oriented Programming. Code ko objects (properties + methods) mein organize karna.	"Car" object: properties (color, speed), methods (start(), stop()).

13. Databases (Daten Ka Kosh)

Database: Data ka organized collection. Jaise ek digital file cabinet. **DBMS**

(Database Management System): Database ko manage karne wala software

(e.g., MySQL, Oracle). **RDBMS (Relational DBMS)**: Data tables mein store hota hai, tables aapas mein related hoti hain.

13.1 Key Terms

Term	Meaning	Analogy (School Marksheet)
Table	Data rows aur columns mein store hota hai.	Marksheet.
Row (Record)	Ek complete entry.	Ek student ka poora data (Roll no, name, marks).
Column (Field)	Ek particular type ki information.	Sabhi students ke "Naam".
Primary Key	Har row ki unique pehchan. Duplicate nahi ho sakta, null nahi ho sakta.	Roll Number.
Foreign Key	Doosri table ki Primary Key ka reference, relationship banane ke liye.	Fee receipt mein Roll Number.
SQL	Structured Query Language. Database se baat karne ki bhasha.	<pre>SELECT * FROM Students WHERE City='Delhi';</pre>
Index	Table ke ek column par fast search ke liye index banana.	Kitaab ke peeche Index page jaisa.

✅ Topics 11, 12, 13: Summary

Field	Key Concept	Memory Trick
Software	System vs Application	System=Worker, App=Tool.
Translators	Compiler vs Interpreter	Compiler=Poori kitaab, Interpreter=Ek line.
Algorithm	Step-by-step logic	Chai ki recipe.

RDBMS	Tables, PK (Unique), FK (Reference)	PK=Aadhar No., FK=Aadhar in Bank A/c.
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14. Cyber Security (Computer Suraksha)

Aapka data aur system surakshit rakhne ke tarike.

14.1 Malware (Bad Software)

Malware Type	Kaam (Behaviour)	Analogy
Virus	Kisi file ke saath chipakta hai, failta hai jab file share ho.	Sardi-Zukam. Ek se doosre ko.
Worm	Bina kisi file ke apne aap network se failta hai.	Machhar. Active spread.
Trojan	Ek genuine software ki tarah dikhta hai, lekin andar se nuksaan karta hai (backdoor).	Trojan Horse (Lakdi ka ghoda). Gift lagta hai, andar sipahi.
Ransomware	Data ko lock (encrypt) kar deta hai aur unlock karne ke liye paise (ransom) maangta hai.	Kidnapper.
Spyware	Chupke se aapki activities record karta hai (keystrokes, browsing).	Jasoos.
Adware	Gair-zaroori pop-up ads dikhata hai.	Jabardasti ka salesman.

14.2 Defence Mechanisms

- **Antivirus:** Virus/Worm detect karta hai, rokta hai, hatata hai.
- **Firewall:** Network traffic monitor (Allowed / Blocked).
- **Encryption:** Data ko aisi coded language mein badalna jo bina key ke kisi ke samajh na aaye. (Plaintext -> Ciphertext).
- **Hashing:** Data ki ek fixed-size unique digital fingerprint banana. Password store karne ke liye use hota hai. (One-way, wapas nahi ja sakte).
- **Authentication:** Aap wahi ho jo keh rahe ho? (User ID + Password).

- **Authorization:** Aapko iski permission hai? (Aap room mein enter hue, lekin VIP room ki chabhi nahi hai).
- **MFA (Multi-Factor Authentication):** Do ya zyada tarah ki pehchan. Jaise, Password (kuch aap jaante hain) + Mobile OTP (kuch aapke paas hai).
- **Digital Signature:** Ye prove karta hai ki document bhejne wale ne hi banaya hai, aur beech mein badla nahi gaya. Hashing + Public Key se banta hai.
- **Public/Private Key:** Ek lock (Public Key) sabko bhejo, lekin chabhi (Private Key) sirf aapke paas. Koi bhi lock laga kar aapko secret message bhej sakta hai.
- **Phishing:** Fake email ya website banakar aapki sensitive info (password, credit card) churana. Jaise asli bank jaisa dikhne wala fake SMS.
- **Social Engineering:** Technology ko hack nahi karte, **insaan ko manipulate karte hain**. Jaise, IT admin ban ke call karke password puchh lena.

✓ Topic 14: Summary

Threat/Defence	One-Liner
Virus vs Worm	File attach (Sardi) vs Auto spread (Machhar).
Encryption	Data lock (Cipher).
Hashing	Data ki fingerprint. One-way.
MFA	2+ methods for login. (Pass + OTP).
Phishing	Fake bait for info.

15. Modern Technologies (Aadhunik Takniki)

15.1 AI, ML, DL (Dimaag Ki Nakal)

- **AI (Artificial Intelligence):** Machine mein insaano jaisi soch. (Broad field).
- **ML (Machine Learning):** AI ki ek technique jisme machine data se seekhti hai, bina explicitly program kiye. Jaise, hazaaron pictures dekh kar "Billi" pehchanna.
- **Deep Learning:** ML ke andar Neural Networks use karke brain ki aur bhi gehri nakal. Image Recognition, Natural Language mein best.
- **Generative AI:** Naya content create karna. Text, images, code. (Jaise ChatGPT, Midjourney).

15.2 Cloud, Virtualization, Containers

- **Cloud Computing:** Internet par servers use karna (Digital Ocean/AWS) bina khud ke physical servers kharide.
- **Virtualization:** Ek physical server ko software se kai chhote virtual servers mein baantna. Resource bachat.
- **Containers (Docker):** Application aur uski dependencies ko ek box mein pack karna jo kisi bhi system par chal sake. Virtualization se halka. Jaise shipping container.
- **Kubernetes:** Harzon containers ko manage karne wala automatic manager.

15.3 Blockchain, IoT, Quantum

- **Blockchain:** Ek digital ledger jo kai computers par bikhra hota hai. Change karna impossible. **Cryptocurrency (Bitcoin)** iska use hai.
- **IoT (Internet of Things):** Rozana ki cheezein (Fan, Light, Fridge) internet se jud jaayen. Sensor + Internet. Smart Home.
- **Edge Computing:** Data ko cloud ke paas bhejne ki jagah, jahan data create ho raha hai (edge) wahin process karo. Latency kam.
- **Quantum Computing:** 0 aur 1 ki jagah **Qubits**. Ek Qubit ek saath 0 aur 1 ho sakta hai (Superposition). Bahut tez. Abhi experimental.

16. Computer Performance (Computer Ki Raftaar)

16.1 Key Terms

Term	Meaning
GHz	Processor ki speed ka unit (higher = faster).
Core	CPU ke andar ek independent processor. Dual-core = 2, Quad-core = 4.
Thread	Core ke andar logical worker. Multithreading mein ek core do threads sambhal sakta hai.
Cache (L1, L2, L3)	CPU ke paas ki super-fast memory. L1 (chhoti, tez, per core), L2 (badhi, per core ya shared), L3 (sabse badi, sab cores share).

Benchmark	Performance score comparison test.
Bottleneck	Kisi ek weak component ki wajah se poora system slow ho jaana. (Jaise F1 car mein cycle ka tyre).
Overclocking	CPU/GPU ki speed factory limit se zyada badhana, risk hai, heat zyada hoti hai.
Thermal Throttling	Zyada garam hone par safety ke liye CPU ka khud speed kam kar dena.

17. Troubleshooting (Samadhan)

Jab computer theek se kaam na kare to aap kya karenge, iski ABCD.

17.1 Common Problems & Quick Fixes

1. Slow PC:

- *Check:* Task Manager (Ctrl+Shift+Esc) kholkar dekho kaunsa app CPU/RAM 100% kar raha hai.
- *Fix:* Startup programs band karo, SSD lagao, RAM badhao.

2. Blue Screen of Death (BSOD):

- *Cause:* Hardware failure ya driver conflict.
- *Fix:* Error code note karo (jaise `IRQL_NOT_LESS_OR_EQUAL`), Google karo, driver update karo.

3. Boot Failure (Black Screen):

- *Check:* Kya monitor cable theek se lagi hai? Kya koi beep sound a rahi hai? (Beep codes ka matlab motherboard manual mein hota hai).
- *Fix:* BIOS setting reset karo (CMOS battery nikal kar wapas daalo), ya Windows Repair Tool use karo.

4. Internet Problems:

- *Check:* Kya Wi-Fi on hai? Kya doosre devices mein chal raha hai internet?
- *Fix:* Router restart karo (10 sec power off). Windows Network Troubleshooter chalao.

5. Safe Mode:

- *Kya Hai:* Windows ka minimal version. Sirf zaroori drivers load hote

hain.

- *Use Case:* Agar koi virus ya driver problem se PC crash ho raha hai, to Safe Mode mein jaakar problem fix kar sakte hain.

A to Z Syllabus Completed! Final Maha-Revision

Shivani, aapne zero se expert tak ka safar tay kar liya hai. Ye aakhri summary hai, jise aap baar-baar padh sakti hain:

Chapter	Sabse Important Points (Keywords)
1. Basics	Babbage, ENIAC, 5 Generations (V-T-I-V-U), GIGO, Diligence.
2. Number System	Decimal(10), Binary(2), Octal(8), Hex(16). ASCII 'A'=65.
3. Data Rep.	Bit -> Byte(8) -> KB(1024) -> MB -> GB -> TB.
4. Hardware	CPU=ALU+CU+Reg. RAM(Volatile, Table) vs ROM(Non-Vol, Shelf). BIOS/UEFI.
5. Storage	HDD(Platter, Sasta) vs SSD(Flash, Tez). NVMe > SATA. RAID 0(Speed) 1(Safety).
6. Input	Keyboard, Mouse, Touchscreen (I/O both).
7. Output	OLED(Pure Black), Laser(Toner, Office), Inkjet(Liquid, Photo).
8. OS	Kernel(Heart), Shell(Face), Process, Thread, Virtual Memory.
9. Networking	Hub(All) -> Switch(MAC) -> Router(IP). TCP(Reliable) vs UDP(Fast). DNS, HTTP/3 (QUIC).
10. Internet	URL, Domain, Browser, Cookies, Cache.
11. Software	System(OS) vs Application(MS Word). Compiler(Full) vs Interpreter(Line).
12. Programming	Algorithm, Variable, Loop, Array, OOP.
13. DB	Table Row PK FK SQL

	Basic, New, High, SQL
14. Security	Virus/Worm/Trojan, Encryption/Hashing, MFA, Phishing.
15. Modern Tech	AI/ML/DL, Cloud, Docker/K8s, IoT, Quantum.
16. Performance	Cores, Threads, L1/L2/L3 Cache, Bottleneck.
17. Troubleshoot	Task Manager, Safe Mode, Reboot Router.